

硬體描述語言與邏輯合成實驗作業 - HW10

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* 主程式碼、測試程式碼皆需要截圖到作業上，波型圖需截圖

* 需用 pdf 檔繳交

* 請務必按照題目之設定撰寫程式

* 所有題目的 FSM 狀態皆須顯示文字

Q1. Praking Lot

題目說明：

1. 請依下列的硬體架構圖(Fig. 1-1)去設計電路

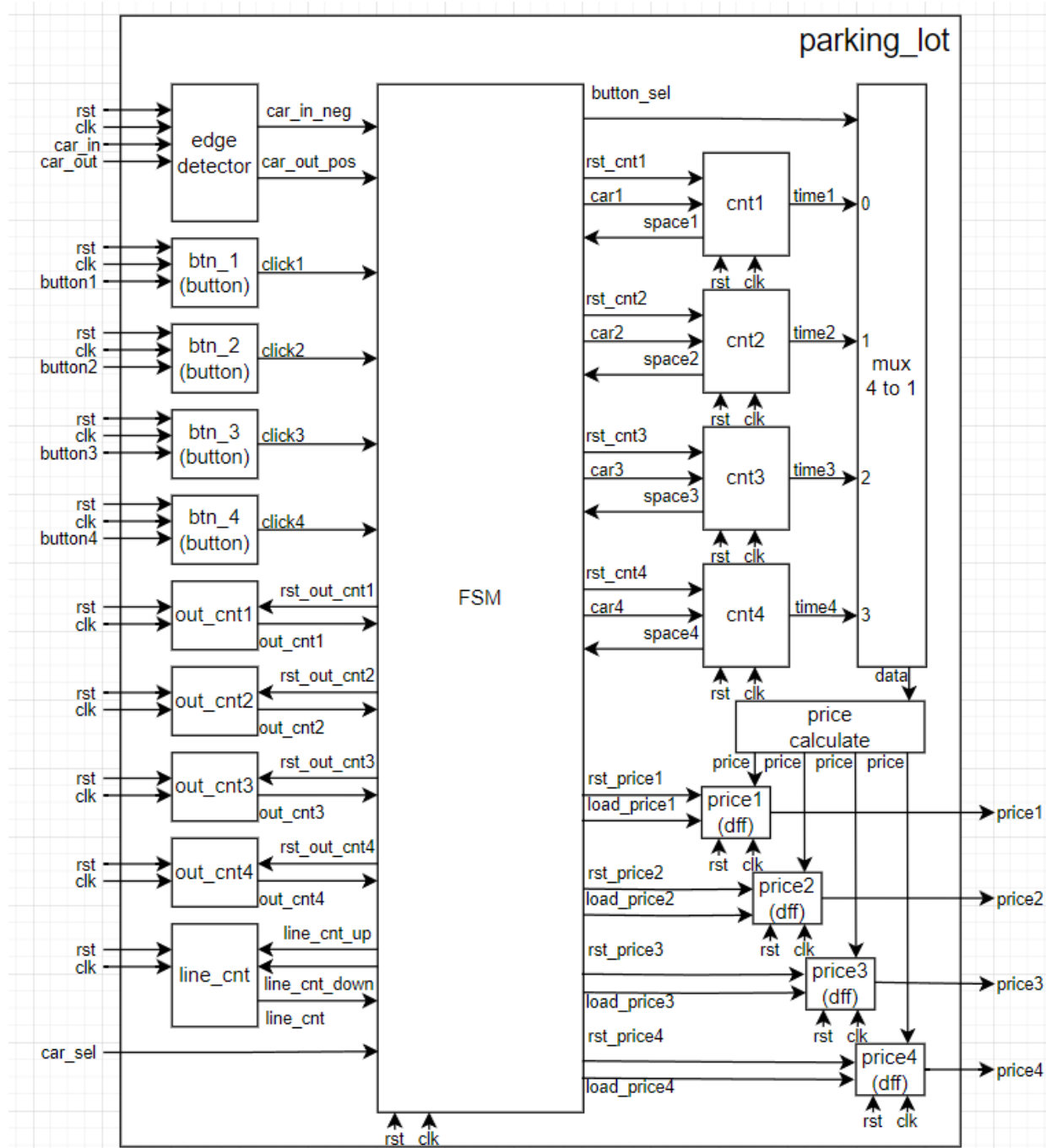


Fig. 1-1 硬體架構圖

2. Input:

- a. clk : system clock
- b. rst : system reset
- c. car_in : 表示停車場有車輛駛入
- d. car_out : 表示停車場有車輛駛出
- e. car_sel[1:0] : 表示停車場從第幾號停車位駛出
- f. botton1、botton2、botton3、botton4 : 表示相對應編號車格內車子進行繳費

Output:

- a. price1 [7:0] : 表示 1 號車格內車輛離場時的價格計費
- b. price2 [7:0] : 表示 2 號車格內車輛離場時的價格計費
- c. price3 [7:0] : 表示 3 號車格內車輛離場時的價格計費
- d. price4 [7:0] : 表示 4 號車格內車輛離場時的價格計費

3. 功能說明:

- a. 當車輛駛入停車場時，對 car_in 進行負緣偵測
- b. 對於駛入的車輛進行車位分配，優先順序如下:

space1 >> space2 >> space3 >> space4

若該 space 無停放車輛則分配該 space 給車輛，並開始進行計時

- c. 當車輛離開停車場時，對 car_out 進行正緣偵測
- d. 若有兩車輛同時駛入及駛出停車場，則以駛出者為優先判斷
- e. 若車位已停滿則進行排隊(line_cnt [2:0])
- f. line_cnt 說明: (line_cnt [2:0])

表示在停車場入口處，排隊的車輛數目，須能夠上數及下數

當有 car_in 訊號觸發，且車位已停滿時 line_cnt + 1 (line_cnt_up == 1)

當有 car_out 訊號時 line_cnt - 1 (line_cnt_down == 1)

- g. cnt 內含的資訊說明如下:

- 1. space : 在 counter[6]，表示為停放車輛的有無(有停放車輛: 1 / 無停放車輛: 0)
- 2. sel_rate: 在 counter[5:4]，表示停車的計費費率
- 3. time: 在 counter[5:0]，表示停車的停放時長
- 4. 當 rst 或是 rst_cnt 為 1 時將該 cnt 設置為 7'b0_00_0000
- 5. 當 car 為 1 時將該 cnt 設置為 7'b1_00_0000
- 6. 其餘時候 cnt = cnt + 7'd1

- h. data 內含的資訊說明如下:

- 1. price_sel: 在 data[5:4]，表示駛出車輛停車費率的選擇
- 2. price_time: 在 data[5:0]，表示駛出車輛的停放時間

- i. price_calculate 說明如下: (請注意該部分禁止使用乘法器去做運算)

- 1. 若 price_sel 為 2'b00 時，price = price_time
- 2. 若 price_sel 為 2'b01 時，price = 16 + (price_time[3:0]*2)
- 3. 若 price_sel 為 2'b10 時，price = 16 + 16*2 + (price_time[3:0]*4)
- 4. 若 price_sel 為 2'b11 時，price = 16 + 16*2 + 16*4 + (price_time[3:0]*8)

- j. price_dff 說明:

當 click 訊號為 1 時，load_price = 1，將 price_calculate 數值儲存

k. out_cnt 說明:

計時 click 訊號為 1 到車輛真實離場間的時間

當(out_cnt > 9)，則 load_price = 1，將原本 price_dff 裡的數值以新的值取代

l. btn1~4 說明:

除彈跳，去除 button 訊號的雜訊，輸出無干擾的訊號 click

m. FSM 的狀態說明如下:

1. IDLE: 起始狀態無條件進入狀態 STAY

2. STAY: 等待車輛駛入或是車輛駛出

a. 當車輛駛入時進入狀態 IN

b. 當車輛駛出時進入狀態 OUT

c. 當任一 click 訊號為 1 時進入狀態 PAY

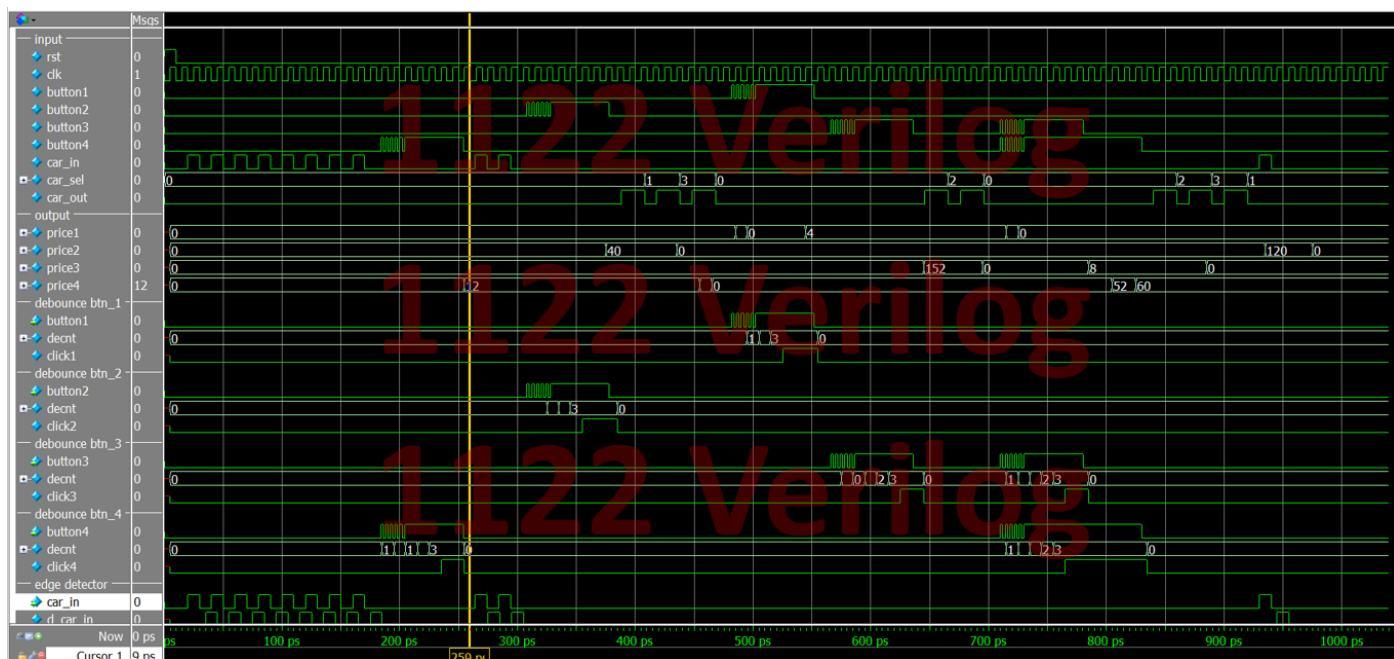
3. IN: 車輛駛入時為其分配車位並將該車位的 car 訊號拉為 1，並將 price_dff 數值規 0，若停車場無空位 line_cnt_up = 1，將排隊數量加 1

4. OUT: 車輛駛出時將該車位 cnt 的 rst_cnt 拉為 1，若超過離場時間，則選擇該車位的 button_sel 並重新 load_price，若 line_cnt 不為 0 時則進入到狀態 IN，line_cnt 減 1

5. PAY: 查看何種 click 為 1，將該對應的 load_price 載入，並重置對應的 out_cnt，並設定其對應的 button_sel，若有同時多個 click 為 1 時則優先順序如下:

click1 >> click 2 >> click 3 >> click 4

波型圖如下 (Fig. 1-2):



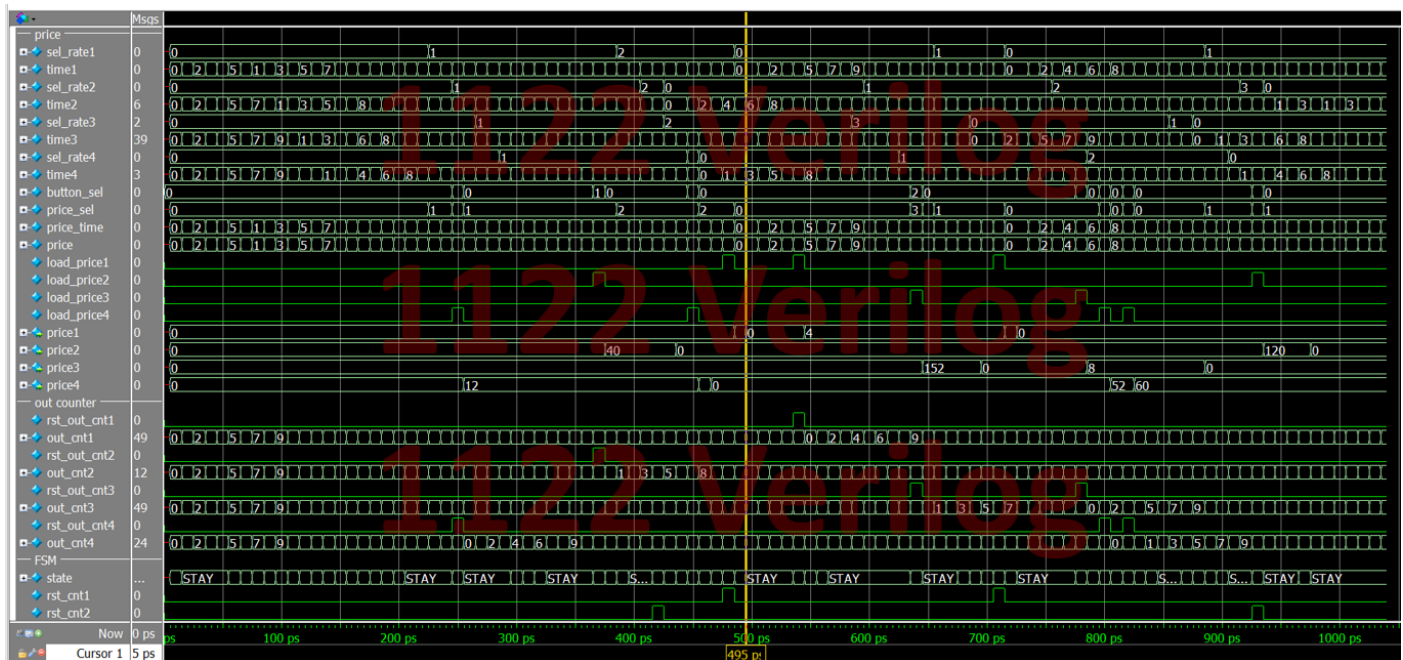
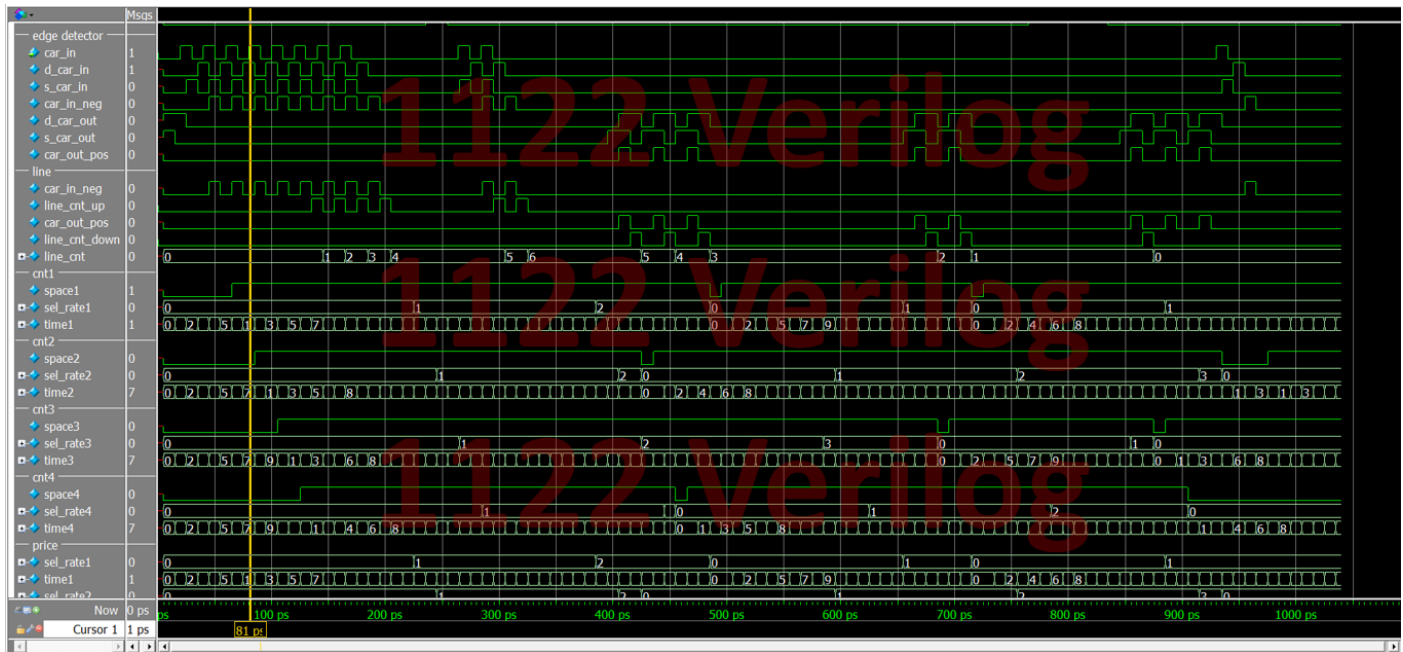


Fig. 1-2 波型圖

主程式碼截圖：

parking_lot.v

//放主程式碼截圖

```
1 module parking_lot(  
2     input clk,rst,car_in,car_out,  
3     input [1:0]car_sel,  
4     input button1,button2,button3,button4,  
5     output reg [7:0]price1,price2,price3,price4  
6 );  
7     reg s_car_out,d_car_out,car_out_pos,s_car_in,d_car_in,car_in_neg;  
8     reg car1,car2,car3,car4;  
9     reg rst_cnt1,rst_cnt2,rst_cnt3,rst_cnt4;  
10    reg load_price1,load_price2,load_price3,load_price4,rst_price1,rst_price2,rst_price3,rst_price4;  
11    reg line_cnt_up,line_cnt_down;  
12    reg rst_out_cnt1,rst_out_cnt2,rst_out_cnt3,rst_out_cnt4;  
13    reg [31:0]state,ns;  
14    reg [7:0]price;  
15    reg [6:0]cnt1,cnt2,cnt3,cnt4;  
16    reg [5:0]out_cnt1,out_cnt2,out_cnt3,out_cnt4;  
17    reg [2:0]line_cnt;  
18    reg [5:0]mux_out,data;  
19    reg [1:0]button_sel;  
20    wire [5:0]time1,time2,time3,time4,price_time;  
21    wire [1:0]sel_rate1,sel_rate2,sel_rate3,sel_rate4,price_sel;  
22    wire space1,space2,space3,space4;  
23    wire click1,click2,click3,click4;  
24  
25    assign space1=cnt1[6];  
26    assign space2=cnt2[6];  
27    assign space3=cnt3[6];  
28    assign space4=cnt4[6];  
29    assign sel_rate1=cnt1[5:4];  
30    assign sel_rate2=cnt2[5:4];  
31    assign sel_rate3=cnt3[5:4];  
32    assign sel_rate4=cnt4[5:4];  
33    assign time1=cnt1[5:0];  
34    assign time2=cnt2[5:0];  
35    assign time3=cnt3[5:0];  
36    assign time4=cnt4[5:0];  
37    assign price_sel=data[5:4];  
38    assign price_time=data[5:0];  
39  
40    always @(posedge clk)  
41        if(rst)
```



```

41     if(rst)
42     begin
43         s_car_out<=1;
44         d_car_out<=1;
45         car_out_pos<=0;
46         s_car_in<=0;
47         d_car_in<=0;
48         car_in_neg<=0;
49     end
50     else
51     begin
52         {d_car_out,s_car_out}<={s_car_out,car_out};
53         car_out_pos<=s_car_out&~d_car_out;
54         {d_car_in,s_car_in}<={s_car_in,car_in};
55         car_in_neg<=~s_car_in&d_car_in;
56     end
57     always @(posedge clk)
58     if(rst)
59         line_cnt<=3'b0;
60     else if(line_cnt_up)
61         line_cnt<=line_cnt+3'd1;
62     else if(line_cnt_down)
63         line_cnt<=line_cnt-3'd1;
64     else
65         line_cnt<=line_cnt;
66
67     always @(posedge clk)
68         if (rst||rst_cnt1)
69             cnt1<=7'b0;
70         else if (car1)
71             cnt1<=7'b1000000;
72         else
73             cnt1<=cnt1+7'd1;
74     always @(posedge clk)
75         if (rst||rst_cnt2)
76             cnt2<=7'b0;
77         else if (car2)
78             cnt2<=7'b1000000;
79         else
80             cnt2<=cnt2+7'd1;
81     always @(posedge clk)

```

```
81  always @(posedge clk)
82      if(rst||rst_cnt3)
83          cnt3<=7'b0;
84      else if (car3)
85          cnt3<=7'b1000000;
86      else
87          cnt3<=cnt3+7'd1;
88  always @(posedge clk)
89      if(rst||rst_cnt4)
90          cnt4<=7'b0;
91      else if (car4)
92          cnt4<=7'b1000000;
93      else
94          cnt4<=cnt4+7'd1;
95
96  always @(posedge clk)
97      if(rst||rst_price1)
98          price1<=8'd0;
99      else if(load_price1)
100         price1<=price;
101      else
102         price1<=price1;
103  always @(posedge clk)
104      if(rst||rst_price2)
105         price2<=8'd0;
106      else if(load_price2)
107         price2<=price;
108      else
109         price2<=price2;
110  always @(posedge clk)
111      if(rst||rst_price3)
112         price3<=8'd0;
113      else if(load_price3)
114         price3<=price;
115      else
116         price3<=price3;
117  always @(posedge clk)
118      if(rst||rst_price4)
119         price4<=8'd0;
120      else if(load_price4)
121         price4<=price;
```



```

121         price4<=price;
122     else
123         price4<=price4;
124
125     always @(posedge clk)
126         if(rst||rst_out_cnt1)
127             out_cnt1<=6'd0;
128         else
129             out_cnt1<=out_cnt1+6'd1;
130     always @(posedge clk)
131         if(rst||rst_out_cnt2)
132             out_cnt2<=6'd0;
133         else
134             out_cnt2<=out_cnt2+6'd1;
135     always @(posedge clk)
136         if(rst||rst_out_cnt3)
137             out_cnt3<=6'd0;
138         else
139             out_cnt3<=out_cnt3+6'd1;
140     always @(posedge clk)
141         if(rst||rst_out_cnt4)
142             out_cnt4<=6'd0;
143         else
144             out_cnt4<=out_cnt4+6'd1;
145
146 module button(
147     output reg    click,
148     input          in,
149     input          clk,
150     input          rst
151 );
152     reg    [1:0]    decnt;
153     parameter bound = 2'd3;
154     always @(posedge clk) begin
155         if(rst) begin
156             decnt    <= 2'd0;
157             click    <= 1'b0;
158         end
159         else if(in) begin
160             if(decnt == bound) begin
161                 decnt    <= decnt;

```

```

161         decnt    <= decnt;
162         click    <= 1'b1;
163     end
164     else begin
165         decnt    <= decnt + 2'd1;
166         click    <= 1'b0;
167     end
168 end
169 else begin
170     decnt    <= 2'd0;
171     click    <= 1'b0;
172 end
173 end
174 endmodule

175
176 button btn_1(click1,button1,clk,rst);
177 button btn_2(click2,button2,clk,rst);
178 button btn_3(click3,button3,clk,rst);
179 button btn_4(click4,button4,clk,rst);
180
181 always @(*)
182 begin
183     case(button_sel)
184     2'b00:data=time1;
185     2'b01:data=time2;
186     2'b10:data=time3;
187     2'b11:data=time4;
188     endcase
189 end
190 always @(*)
191 begin
192     case(price_sel)
193     2'b00:price={2'd0,price_time};
194     2'b01:price=8'd16 + ({4'd0,price_time[3:0]}<<1);
195     2'b10:price=8'd48 + ({4'd0,price_time[3:0]}<<2);
196     2'b11:price=8'd112 + ({4'd0,price_time[3:0]}<<3);
197     endcase
198 end
199 always @(posedge clk)
200 begin
201     if(rst)

```

```

201         if(rst)
202             state<="IDLE";
203         else
204             state<=ns;
205     end
206     always @(posedge rst)
207     begin
208         car1=0;
209         car2=0;
210         car3=0;
211         car4=0;
212         rst_cnt1=0;
213         rst_cnt2=0;
214         rst_cnt3=0;
215         rst_cnt4=0;
216         rst_out_cnt1=0;
217         rst_out_cnt2=0;
218         rst_out_cnt3=0;
219         rst_out_cnt4=0;
220         load_price1=0;
221         load_price2=0;
222         load_price3=0;
223         load_price4=0;
224         rst_price1=0;
225         rst_price2=0;
226         rst_price3=0;
227         rst_price4=0;
228         line_cnt_up=0;
229         line_cnt_down=0;
230         button_sel=2'd0;
231     end
232     always @(*)
233     begin
234         case(state)
235             "IDLE":
236             begin
237                 car1=0;
238                 car2=0;
239                 car3=0;
240                 car4=0;
241                 rst_cnt1=0;

```

```
241     rst_cnt1=0;
242     rst_cnt2=0;
243     rst_cnt3=0;
244     rst_cnt4=0;
245     rst_out_cnt1=0;
246     rst_out_cnt2=0;
247     rst_out_cnt3=0;
248     rst_out_cnt4=0;
249     load_price1=0;
250     load_price2=0;
251     load_price3=0;
252     load_price4=0;
253     rst_price1=0;
254     rst_price2=0;
255     rst_price3=0;
256     rst_price4=0;
257     line_cnt_up=0;
258     line_cnt_down=0;
259     button_sel=2'd0;
260     ns="STAY";
261 end
262 "STAY":
263 begin
264     car1=0;
265     car2=0;
266     car3=0;
267     car4=0;
268     rst_cnt1=0;
269     rst_cnt2=0;
270     rst_cnt3=0;
271     rst_cnt4=0;
272     rst_out_cnt1=0;
273     rst_out_cnt2=0;
274     rst_out_cnt3=0;
275     rst_out_cnt4=0;
276     load_price1=0;
277     load_price2=0;
278     load_price3=0;
279     load_price4=0;
280     rst_price1=0;
281     rst_price2=0;
```

```

281 rst_price2=0;
282 rst_price3=0;
283 rst_price4=0;
284 line_cnt_up=0;
285 line_cnt_down=0;
286 button_sel=2'd0;
287 if(car_in_neg)
288     ns="IN";
289 else if(car_out_pos)
290     ns="OUT";
291 else if(click1||click2||click3||click4)
292     ns="PAY";
293 else
294     ns="STAY";
295 end
296 "IN":
297 begin
298     rst_cnt1=0;
299     rst_cnt2=0;
300     rst_cnt3=0;
301     rst_cnt4=0;
302     load_price1=0;
303     load_price2=0;
304     load_price3=0;
305     load_price4=0;
306     line_cnt_down=0;
307     button_sel=2'd0;
308     if(space1==0)
309     begin
310         ns="STAY";
311         car1=1;
312         rst_price1=1;
313     end
314     else if(space2==0)
315     begin
316         ns="STAY";
317         car2=1;
318         rst_price2=1;
319     end
320     else if(space3==0)
321     begin

```

```

321      begin
322          ns="STAY";
323          car3=1;
324          rst_price3=1;
325      end
326      else if(space4==0)
327      begin
328          ns="STAY";
329          car4=1;
330          rst_price4=1;
331      end
332      else
333      begin
334          ns="STAY";
335          line_cnt_up=1;
336      end
337  end
338  "OUT":
339  begin
340      line_cnt_up=0;
341      case(car_sel)
342      2'd0:
343      begin
344          rst_cnt1=1;
345          if(out_cnt1>6'd9)
346          begin
347              load_price1=1;
348              button_sel=2'd0;
349          end
350          else
351              load_price1=0;
352          end
353      2'd1:
354      begin
355          rst_cnt2=1;
356          if(out_cnt2>6'd9)
357          begin
358              load_price2=1;
359              button_sel=2'd1;
360          end
361      else

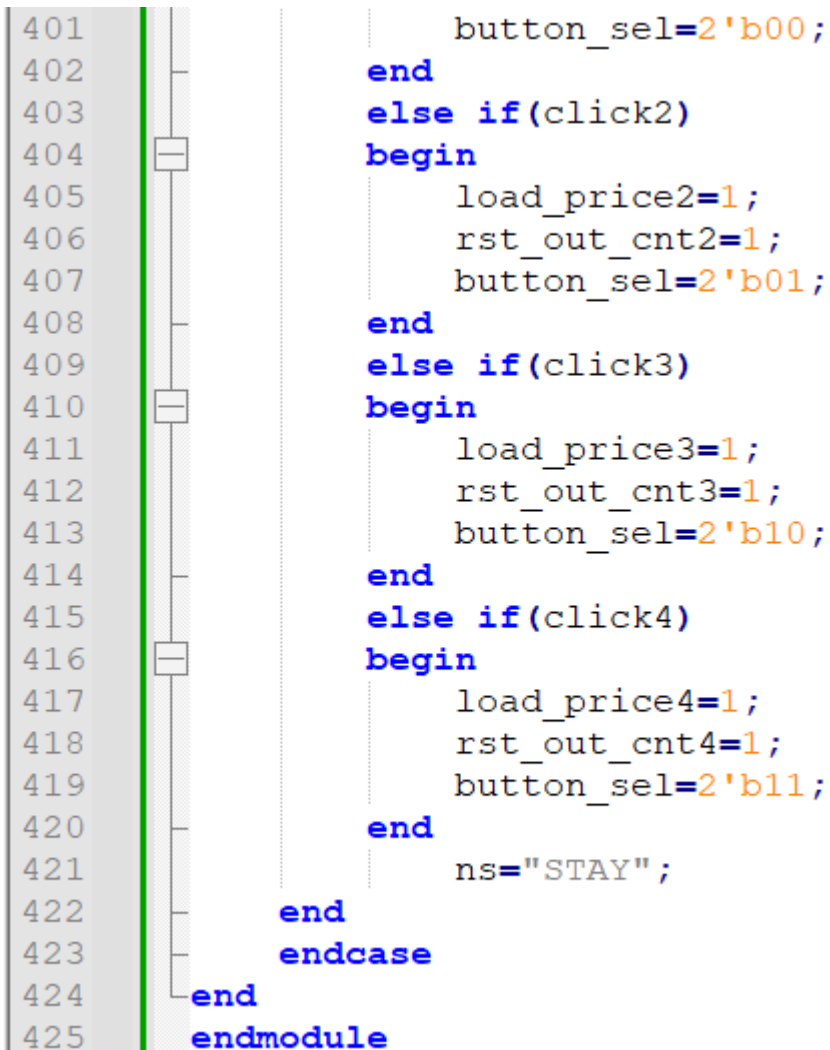
```



```

361         else
362             load_price2=0;
363         end
364         2'd2:
365         begin
366             rst_cnt3=1;
367             if(out_cnt3>6'd9)
368                 begin
369                     load_price3=1;
370                     button_sel=2'd2;
371                 end
372             else
373                 load_price3=0;
374             end
375         2'd3:
376         begin
377             rst_cnt4=1;
378             if(out_cnt4>6'd9)
379                 begin
380                     load_price4=1;
381                     button_sel=2'd3;
382                 end
383             else
384                 load_price4=0;
385             end
386         endcase
387         if(line_cnt>6'd0)
388         begin
389             line_cnt_down=1;
390             ns="IN";
391         end
392         else
393             ns="STAY";
394     end
395     "PAY":
396     begin
397         if(click1)
398         begin
399             load_price1=1;
400             rst_out_cnt1=1;
401             button_sel=2'b00;

```



```
401         button_sel=2'b00;
402     end
403     else if(click2)
404     begin
405         load_price2=1;
406         rst_out_cnt2=1;
407         button_sel=2'b01;
408     end
409     else if(click3)
410     begin
411         load_price3=1;
412         rst_out_cnt3=1;
413         button_sel=2'b10;
414     end
415     else if(click4)
416     begin
417         load_price4=1;
418         rst_out_cnt4=1;
419         button_sel=2'b11;
420     end
421     ns="STAY";
422 end
423 endcase
424 end
425 endmodule
```

測試程式碼截圖：

parking_lot_tb.v

//放測試程式碼截圖

```

427 module parking_lot_tb;
428     wire [7:0] price1;
429     wire [7:0] price2;
430     wire [7:0] price3;
431     wire [7:0] price4;
432     reg  [1:0] car_sel;
433     reg      car_in;
434     reg      car_out;
435     reg      button1;
436     reg      button2;
437     reg      button3;
438     reg      button4;
439     reg      clk;
440     reg      rst;
441     integer   i;
442
443     parking_lot parking_lot_1(
444         .price1      (price1 ),
445         .price2      (price2 ),
446         .price3      (price3 ),
447         .price4      (price4 ),
448         .car_sel     (car_sel),
449         .car_in      (car_in ),
450         .car_out     (car_out),
451         .button1     (button1),
452         .button2     (button2),
453         .button3     (button3),
454         .button4     (button4),
455         .clk         (clk    ),
456         .rst         (rst    )
457     );
458
459     always #5 clk = ~clk;
460
461     task drive_in;
462     begin
463         #10 car_in = 1'b1;
464         #10 car_in = 1'b0;
465     end
466 endtask

```

```
461 task drive_in;
462     begin
463         #10 car_in = 1'b1;
464         #10 car_in = 1'b0;
465     end
466 endtask
467 task drive_out;
468     begin
469         #10 car_out = 1'b1;
470         #20 car_out = 1'b0;
471     end
472 endtask
473 task button1_on;
474     begin
475         #10 btn1_noise_on;
476         #50 button1 = 1'b0;
477     end
478 endtask
479 task btn1_noise_on;
480     begin
481         for (i = 0; i < 6; i = i + 1) begin
482             #2 button1 = 1'b0;
483             #2 button1 = 1'b1;
484         end
485     end
486 endtask
487 task button2_on;
488     begin
489         #10 btn2_noise_on;
490         #50 button2 = 1'b0;
491     end
492 endtask
493 task btn2_noise_on;
494     begin
495         for (i = 0; i < 6; i = i + 1) begin
496             #2 button2 = 1'b0;
497             #2 button2 = 1'b1;
498         end
499     end
500 endtask
501 task button3_on;
```

```

496         #2 button2 = 1'b0;
497         #2 button2 = 1'b1;
498     end
499 end
500 endtask
501 task button3_on;
502     begin
503         #10 btn3_noise_on;
504         #50 button3 = 1'b0;
505     end
506 endtask
507 task btn3_noise_on;
508     begin
509         for (i = 0; i < 6; i = i + 1) begin
510             #2 button3 = 1'b0;
511             #2 button3 = 1'b1;
512         end
513     end
514 endtask
515 task button4_on;
516     begin
517         #10 btn4_noise_on;
518         #50 button4 = 1'b0;
519     end
520 endtask
521 task btn4_noise_on;
522     begin
523         for (i = 0; i < 6; i = i + 1) begin
524             #2 button4 = 1'b0;
525             #2 button4 = 1'b1;
526         end
527     end
528 endtask
529 task multi_button_on;
530     begin
531         #10 multi_btn_noise_on;
532         #50 button3 = 1'b0;
533         #50 button4 = 1'b0;
534     end
535 endtask
536 task multi_btn_noise_on;

```

```

536 task multi_btn_noise_on;
537 begin
538     for (i = 0; i < 6; i = i + 1) begin
539         #2 button3 = 1'b0; button4 = 1'b0;
540         #2 button3 = 1'b1; button4 = 1'b1;
541     end
542 end
543 endtask
544 initial begin
545     rst = 1'b1; clk = 1'b0; car_sel = 2'd0; car_in = 1'b0; car_out = 1'b0;
546     button1 = 1'b0; button2 = 1'b0; button3 = 1'b0; button4 = 1'b0;
547     #10 rst = 1'b0;
548     drive_in; // one car in (choose space1)
549     drive_in; // one car in (choose space2)
550     drive_in; // one car in (choose space3)
551     drive_in; // one car in (choose space4)
552     drive_in; // one car in (parking lot full, line_cnt = 1)
553     drive_in; // one car in (parking lot full, line_cnt = 2)
554     drive_in; // one car in (parking lot full, line_cnt = 3)
555     drive_in; // one car in (parking lot full, line_cnt = 4)
556     button4_on; // pay price (space4)
557     drive_in; // one car in (parking lot full, line_cnt = 5)
558     drive_in; // one car in (parking lot full, line_cnt = 6)
559     button2_on; // pay price (space2)
560     drive_out; car_sel = 2'd1; // space2 car out (in out time), one car in(choose space2, line_cnt = 5)
561     drive_out; car_sel = 2'd3; // space4 car out (over out time), one car in(choose space4, line_cnt = 4)
562     drive_out; car_sel = 2'd0; // space1 car out (no pay), one car in(choose space1, line_cnt = 3)
563     button1_on; // pay price (space1)
564     button3_on; // pay price (space3)
565     drive_out; car_sel = 2'd2; // space3 car out (in out time), one car in (choose space3, line_cnt = 2)
566     drive_out; car_sel = 2'd0; // space1 car out (choose space1, line_cnt = 1)
567     multi_button_on; // pay price (space3) first, and then pay price (space4)
568     drive_out; car_sel = 2'd2; // space3 car out (in out time), one car in (choose space3, line_cnt = 0)
569     drive_out; car_sel = 2'd3; // space4 car out (in out time)
570     drive_out; car_sel = 2'd1; // space2 car out (no pay)
571     drive_in; // one car in (choose space2)
572     #100 $stop;
573 end
574 endmodule

```

波形圖截圖：

//放波形圖截圖

